

CampusKey Development Plan

Biometric & RFID Authentication System

3-Week Agile Implementation Plan

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Project Duration: 3 Weeks (15 Working Days)

Methodology: Agile Scrum

Team: Fortune Ogulewe & Kamran Yaghoubian

Technology Stack: Flutter, Python FastAPI, SQLite

Project Overview

CampusKey is a multi-factor authentication system designed to replace vulnerable textual passwords with secure biometric and RFID authentication for Fake University. This document outlines a comprehensive 3-week agile development plan with daily sprints and clear deliverables.

Week 1: Foundation & Core Authentication

Sprint 1: Project Setup & Basic Authentication

Day 1: Project Initialization

- **Both Team Members:**
 - Project repository setup (Git)
 - Development environment configuration
 - VS Code with Flutter & Python extensions
 - Install required dependencies
- **Deliverables:**
 - Git repository with basic structure
 - Working development environments

- Flutter "Hello World" app running on iOS/Android
- Python FastAPI server responding on port 8000

Day 2: Backend API Foundation

- **Both Team Members:**

- Design and implement basic authentication endpoints
- Create user model and database schema
- Implement SQLite database integration
- Set up CORS for mobile app communication

- **Deliverables:**

- Functional /login endpoint
- User registration system
- Database with test users
- API documentation

Day 3: Flutter Authentication UI

- **Both Team Members:**

- Create login screen UI
- Implement form validation
- Design app branding (CampusKey)
- Set up navigation structure

- **Deliverables:**

- Professional login interface
- Form validation working
- App navigation framework
- Consistent branding applied

Day 4: Frontend-Backend Integration

- **Both Team Members:**

- Connect Flutter app to Python backend
- Implement HTTP client services
- Handle API responses and errors
- Test on both iOS and Android devices

- **Deliverables:**

- Successful login from mobile app
- Error handling for network issues
- Loading states and user feedback
- Cross-platform compatibility

Day 5: Sprint Review & Testing

- **Both Team Members:**

- Comprehensive testing on multiple devices
- Bug fixes and performance optimization
- Code review and refactoring
- Sprint retrospective and planning

- **Deliverables:**

- Stable authentication system
- Documentation update
- Week 2 sprint plan
- Test reports from iOS and Android

Week 2: Multi-Factor Authentication Features

Sprint 2: Biometric Integration

Day 6: Biometric Authentication Setup

- **Both Team Members:**

- Research and implement local_auth package
- Configure biometric permissions for iOS/Android
- Design biometric enrollment flow
- Implement secure storage for biometric preferences

- **Deliverables:**

- Biometric authentication working on both platforms
- Face ID/Touch ID integration (iOS)
- Fingerprint scanner integration (Android)
- User preference management

Day 7: Backend Biometric Verification

- **Both Team Members:**

- Extend authentication system for biometric flow
- Implement secure session management
- Add biometric verification endpoints
- Enhance security with token-based authentication

- **Deliverables:**

- Biometric verification API endpoints
- Enhanced session security
- Token refresh mechanism
- Updated API documentation

Day 8: RFID Simulation & Integration

- **Both Team Members:**

- Design RFID simulation system
- Implement NFC reading capabilities
- Create RFID token management
- Develop RFID verification backend

- **Deliverables:**

- RFID simulation working
- NFC reading implementation
- Token verification system
- RFID user association

Day 9: Multi-Factor Flow Integration

- **Both Team Members:**

- Combine biometric and RFID authentication
- Implement multi-factor flow logic
- Create seamless user experience
- Add fallback authentication methods

- **Deliverables:**

- Working multi-factor authentication

- Smooth user flow between factors
- Fallback mechanisms
- Enhanced security logging

Day 10: Sprint Review & Security Enhancement

- **Both Team Members:**

- Security audit and penetration testing
- Performance optimization
- User experience refinement
- Preparation for final sprint

- **Deliverables:**

- Security assessment report
- Performance metrics
- Refined user interface
- Week 3 sprint plan

Week 3: Polish, Testing & Deployment

Sprint 3: Production Ready Implementation

Day 11: Dashboard & User Management

- **Both Team Members:**

- Create user dashboard after authentication
- Implement role-based access control
- Add user profile management
- Develop session management

- **Deliverables:**

- Functional user dashboard
- Role-based UI variations
- Profile management system
- Secure session handling

Day 12: Advanced Security Features

- **Both Team Members:**

- Implement encryption for sensitive data
- Add audit logging system
- Develop anomaly detection
- Enhance error handling and reporting

- **Deliverables:**

- Data encryption implementation
- Comprehensive audit logs
- Security monitoring system
- Enhanced error management

Day 13: Cross-Platform Optimization

- **Both Team Members:**

- iOS-specific optimizations
- Android-specific enhancements
- Performance testing and tuning
- Memory management optimization

- **Deliverables:**

- Optimized iOS performance
- Enhanced Android experience
- Memory leak fixes
- Performance benchmark reports

Day 14: Comprehensive Testing

- **Both Team Members:**

- End-to-end testing on multiple devices
- User acceptance testing
- Security vulnerability assessment
- Performance stress testing

- **Deliverables:**

- Comprehensive test reports

- Bug fix implementation
- Performance optimization complete
- Security validation

Day 15: Final Polish & Documentation

- **Both Team Members:**

- Code cleanup and refactoring
- Final documentation preparation
- Demo preparation and rehearsal
- Project presentation materials

- **Deliverables:**

- Production-ready codebase
- Complete documentation
- Demo application
- Project presentation

Agile Methodology Implementation

Daily Standups (15 minutes)

- **Time:** 9:00 AM Daily

- **Format:**

- What did I accomplish yesterday?
- What will I work on today?
- Any blockers or challenges?

Sprint Reviews (Friday Afternoons)

- Demo completed features
- Review sprint goals
- Gather feedback
- Plan next sprint

Definition of Done

- Code written and reviewed
- Functionality tested on iOS and Android
- Documentation updated
- No critical bugs
- Performance benchmarks met

Risk Management

Technical Risks

- Platform-specific limitations
- Biometric API compatibility
- NFC hardware variations
- Network connectivity issues

Mitigation Strategies

- Early testing on target devices
- Fallback authentication methods
- Comprehensive error handling
- Regular backup and version control

Success Metrics

- Functional authentication on iOS and Android
- Multi-factor authentication working
- Response time under 2 seconds
- Zero critical security vulnerabilities
- Positive user testing feedback